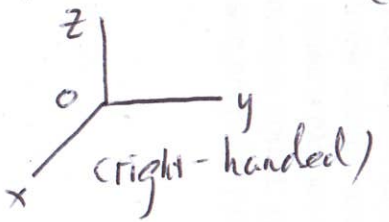
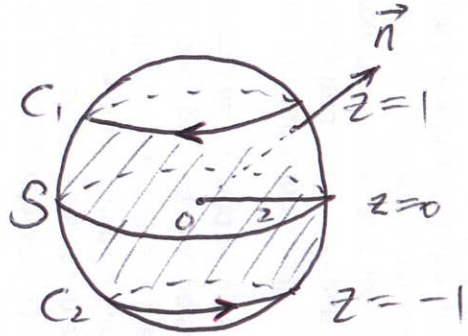
MATH2921 Year 2018

Exam Solution Updates

Q7. Solution: (a)

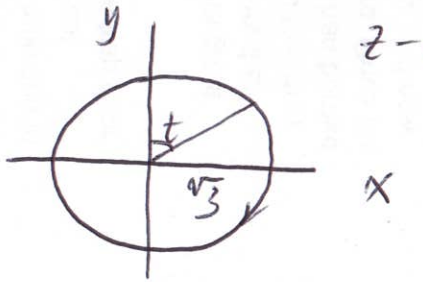




(b) By Stokes' Theorem,

$$\iint_S \text{curl}\,\vec{F}\cdot\vec{n}\,dS = \int_{C_1+C_2} \vec{F}\cdot d\vec{r}.$$

C_1: $z=1$, $x^2+y^2=3$ on the sphere. View from positive z-axis: (above C_1)



$\vec{r}_1(t) = (\sqrt{3}\sin t, \sqrt{3}\cos t, 1)$, $t\in[0,2\pi)$.

$\vec{r}_1'(t) = (\sqrt{3}\cos t, -\sqrt{3}\sin t, 0)$

$\vec{F}(t) = \vec{F}(\vec{r}_1(t)) = \vec{F}(\overset{x(t)}{\sqrt{3}\sin t}, \overset{y(t)}{\sqrt{3}\cos t}, \overset{z(t)}{1})$

$= (\sqrt{3}\cos t + 2\sqrt{3}\sin t, -\sqrt{3}\sin t + 12\cos^2 t, -1)$

$$\int_{C_1} \vec{F}\cdot d\vec{r}_1 = \int_0^{2\pi} (\sqrt{3}\cos t + 2\sqrt{3}\sin t, -\sqrt{3}\sin t + 12\cos^2 t, -1)$$
$$\cdot(\sqrt{3}\cos t, -\sqrt{3}\sin t, 0)\,dt$$

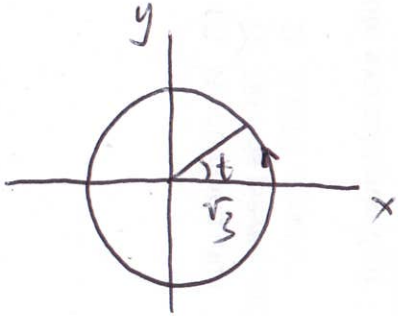
$$= \int_0^{2\pi} (3\cos^2 t + 6\cos t\sin t + 3\sin^2 t - 12\sqrt{3}\cos^2 t\sin t)\,dt$$

$$= \int_0^{2\pi} (3 + 6\cos t\sin t - 12\sqrt{3}\cos^2 t\sin t)\,dt$$

Let's hope for cancellation with $\int_{C_2} \vec{F}\,d\vec{r}$.

C_2: $z=-1$, $x^2+y^2=3$ on the sphere

View from positive z-axis:



$\vec{r}_2(t) = (\sqrt{3}\cos t, \sqrt{3}\sin t, -1)$

$t \in [0, 2\pi)$.

$\vec{r}_2'(t) = (-\sqrt{3}\sin t, \sqrt{3}\cos t, 0)$

$$\vec{F}(t) = \vec{F}(\vec{r}_2(t)) = \vec{F}(\underbrace{\sqrt{3}\cos t}_{x(t)}, \underbrace{\sqrt{3}\sin t}_{y(t)}, \underbrace{-1}_{z(t)})$$

$$= (-\sqrt{3}\sin t + 2\sqrt{3}\cos t,\ \sqrt{3}\cos t + 12\sin^2 t,\ -1)$$

$$\int_{C_2} \vec{F}\cdot d\vec{r}_2 = \int_0^{2\pi} (-\sqrt{3}\sin t + 2\sqrt{3}\cos t,\ \sqrt{3}\cos t + 12\sin^2 t,\ -1) \cdot (-\sqrt{3}\sin t, \sqrt{3}\cos t, 0)\, dt$$

$$= \int_0^{2\pi} (3\sin^2 t - 6\cos t\sin t + 3\cos^2 t + 12\sqrt{3}\sin^2 t\cos t)\, dt$$

$$= \int_0^{2\pi} (3 - 6\cos t\sin t + 12\sqrt{3}\sin^2 t\cos t)\, dt$$

So

$$\int_{C_1+C_2} \vec{F}\cdot d\vec{r} = \int_{C_1} \vec{F}\cdot d\vec{r}_1 + \int_{C_2} \vec{F}\cdot d\vec{r}_2$$

$$= \int_0^{2\pi} (6 - 12\sqrt{3}\cos^2 t\sin t + 12\sqrt{3}\sin^2 t\cos t)\, dt$$

$$= 6\int_0^{2\pi} 1\, dt - 12\sqrt{3}\int_0^{2\pi} \cos^2 t\sin t\, dt + 12\sqrt{3}\int_0^{2\pi} \sin^2 t\cos t\, dt$$

$\cos 2\pi = \cos 0 = 1$

$$= 12\pi + 12\sqrt{3}\int_0^{2\pi} \cancel{\cos^2 t\, d\cos t} + 12\sqrt{3}\int_0^{2\pi} \cancel{\sin^2 t\, d\sin t}$$

$\sin 2\pi = \sin 0 = 0$

$$= 12\pi$$

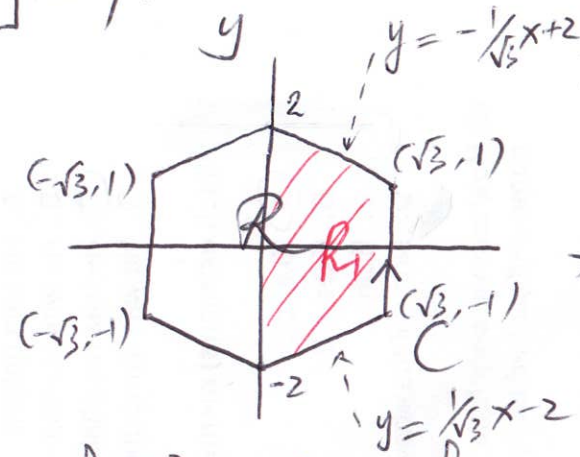
Q8. Solution: the standard method with 4 steps would work.

In the original solution, they use periodic extension and D'Alembert solution which is OK as discussed in Week12-Lecture3-4-PC on Pages 29 — 34.

Of course, the solutions are equivalent.

Q9. Solution:

$F(x,y) = (0, x^3)$



$$\oint_C \vec{F}\cdot d\vec{r} = \oint_C x^3\,dy$$

$$= \iint_R \frac{\partial x^3}{\partial x}\,dx\,dy$$

$$= 3\iint_R x^2\,dx\,dy$$

The right half of R is

$$R_1 = \left\{(x,y) \,\middle|\, 0 \le x \le \sqrt{3},\ \frac{1}{\sqrt{3}}x-2 \le y \le -\frac{1}{\sqrt{3}}x+2\right\}$$

(using a bit symmetry) (*)

$$\oint_C \vec{F}\cdot d\vec{r} \overset{(*)}{=} 6\int_0^{\sqrt{3}}\int_{\frac{1}{\sqrt{3}}x-2}^{-\frac{1}{\sqrt{3}}x+2} x^2\,dy\,dx$$

$$= 6\int_0^{\sqrt{3}} x^2\left(4-\frac{2}{\sqrt{3}}x\right)dx$$

$$= 8x^3\Big|_0^{\sqrt{3}} - \sqrt{3}x^4\Big|_0^{\sqrt{3}}$$

$$= 24\sqrt{3} - 9\sqrt{3}$$

$$= 15\sqrt{3}.$$